Latency Prediction

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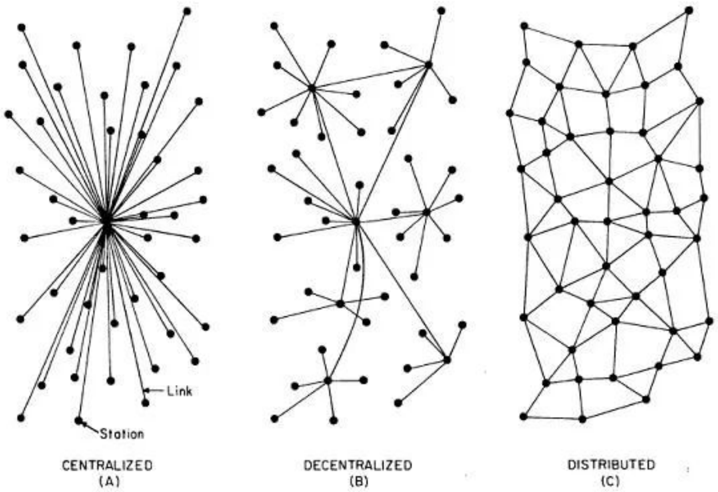
Motivation

- Internet's dominant architecture: Client-Server
- Alternative: Peer-To-Peer
- Protocols govern exchange
- Each of them have tunable hyper parameters based on a global network view
- Tuning of some parameters benefits from knowing the **latency distribution among all nodes**
- Use cases in IoT, WebRTC, IM

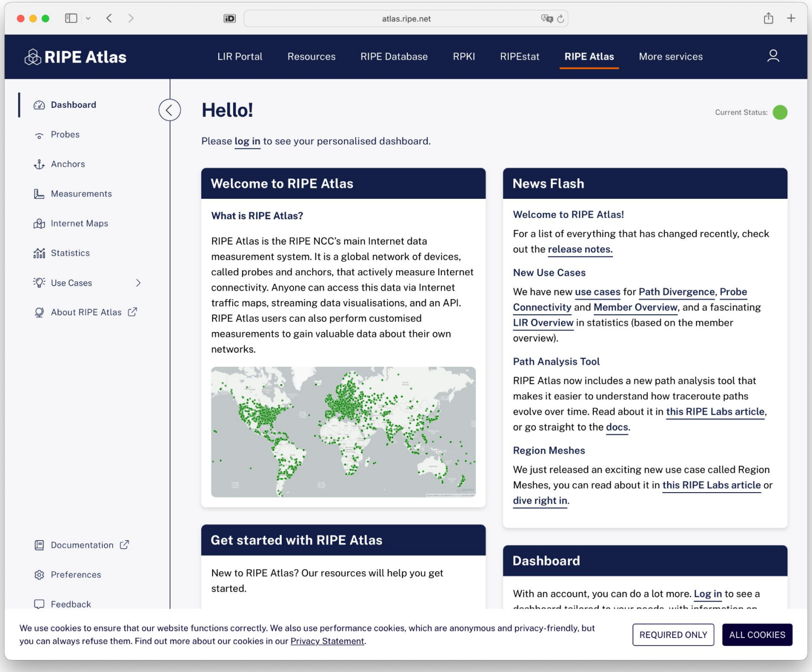


Goal

- We want to predict the latency **between two peers**.
- The catch: in real-life we can never measure peer ↔ peer latency directly
- We only get to measure from known locations → to each peer.
- Our model will have to infer peer↔peer latency from those probe measurements.

Data Sources I - RIPE Atlas

- Globally distributed community-run probes
- Performing various constantly various measurements
- Relevant for us:
  - Ping
  - Traceroute



Data Sources I - RIPE Atlas

- Globally distributed community-run probes
- Performing various constantly various measurements
- Relevant for us:
  - **Ping**
  - Traceroute

	timestamp	src_addr	dst_addr	sent	rcvd	dup	min	max	avg
1	1779064285000000	140.235.236.25	128.164.88.178	3	3	0	31.793486	31.879614	31.84020466666667
2	1779064525000000	103.167.250.30	54.39.185.31	3	3	0	226.826672	227.364399	227.03987266666664
3	1779064325000000	149.62.153.190	213.225.160.239	3	3	0	7.06493	7.161879	7.122001666666667
4	1779064599000000	144.208.197.116	153.92.43.251	3	3	0	32.305197	32.567055	32.44842499999999
5	1779065769000000	144.208.192.250	92.223.59.77	3	3	0	36.45458	36.558756	36.505001
6	1779064374000000	193.17.90.35	193.0.0.164	3	3	0	20.675051	20.737599	20.711742333333333
7	1779064328000000	140.235.236.25	91.221.37.27	3	3	0	173.375866	173.612517	173.49813533333335
8	1779065299000000	102.217.156.154	216.238.41.68	3	3	0	231.73531	231.81406	231.77417733333334
9	1779065319000000	193.17.90.35	212.202.68.1	3	3	0	1.064465	1.206204	1.1581186666666665
10	1779065807000000	185.78.84.3	109.68.162.233	3	3	0	40.215618	40.26181	40.239861
11	1779064383000000	206.196.178.150	82.200.168.242	3	3	0	230.988786	231.227333	231.09437366666666
12	1779063316000000	103.167.250.30	193.17.192.164	3	3	0	192.402841	196.698824	194.53232066666666
13	1779064347000000	193.180.242.2	188.143.128.12	3	3	0	47.888022	48.232238	48.043631
14	1779064575000000	5.39.202.17	62.115.157.74	3	3	0	149.338306	149.456251	149.40283066666666
15	1779064379000000	128.0.113.132	154.197.77.19	3	3	0	292.907842	294.048041	293.4351286666667
16	1779064144000000	114.31.207.7	170.210.5.200	3	3	0	393.111784	393.225897	393.1814583333333

Data Sources I - RIPE Atlas

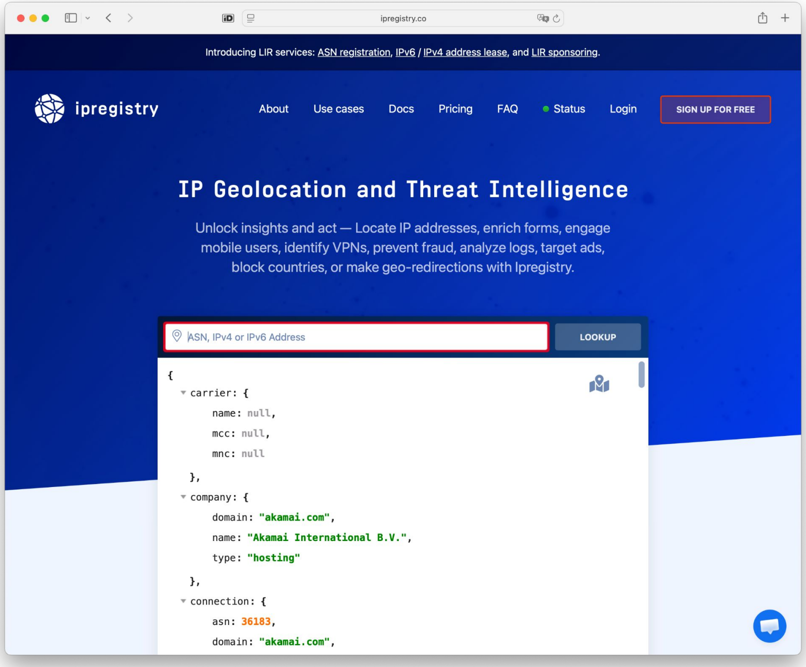
- Globally distributed community-run probes
- Performing various constantly various measurements
- Relevant for us:
  - Ping
  - **Traceroute**

	timestamp	src_addr	dst_addr	hop_count
1	1779062506000000	185.108.68.174	37.10.40.14	10
2	1779065258000000	45.76.86.163	156.154.85.254	9
3	1779063391000000	202.214.97.16	192.50.27.66	10
4	1779062503000000	23.135.52.20	37.10.40.14	13
5	1779064271000000	85.120.32.157	136.0.251.253	27
6	1779063303000000	200.1.122.55	200.94.183.170	14
7	1779063464000000	195.187.36.10	8.8.8.8	6
8	1779064354000000	213.212.129.68	192.54.112.30	6
9	1779062544000000	81.211.198.45	143.215.193.117	13
10	1779063411000000	38.211.235.130	192.58.128.30	11
11	1779062532000000	104.131.160.184	85.120.32.157	9
12	1779064343000000	213.184.84.88	194.187.91.17	12
13	1779062558000000	195.113.165.61	212.10.11.213	9
14	1779063417000000	130.59.80.2	185.17.243.228	8
15	1779063425000000	193.33.184.87	138.185.228.1	17

Data Sources II - IPRegistry

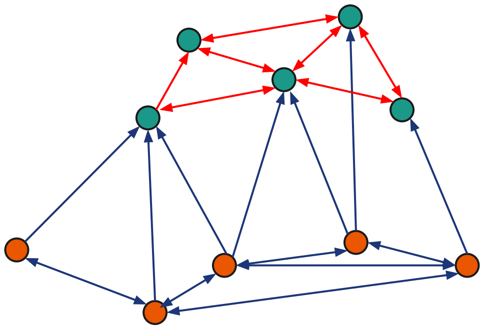
- Gives us geographical information for each IP

```
{
    "ip": pl.String,
    "continent": pl.String,
    "country": pl.String,
    "longitude": pl.Float64,
    "latitude": pl.Float64,
    "asn": pl.Int64,
    "conn_type": pl.String,
    "provider": pl.String,
    "is_mobile": pl.Boolean,
    "is_cloud_provider": pl.Boolean,
    "is_proxy": pl.Boolean,
    "is_vpn": pl.Boolean,
    "is_tor": pl.Boolean,
    "is_tor_exit": pl.Boolean,
}
```



Data Sources III - Final Datasets

- Full dataset: 1.12 M rows
- Split dataset into **landmarks** and **peers**



1_DatasetCharacteristics/full_df.parquet

timestamp	timestamp	src_addr	varchar	dst_addr	varchar
sent	bigint	rcvd	bigint	dup	bigint
min	double	max	double	avg	double
src_continent	varchar	src_country	varchar	src_longitude	double
src_latitude	double	src_asn	bigint	src_conn_type	varchar
src_provider	varchar	src_is_mobile	boolean	src_is_cloud_provider	boolean
src_is_proxy	boolean	src_is_vpn	boolean	src_is_tor	boolean
src_is_tor_exit	boolean	dst_continent	varchar	dst_country	varchar
dst_longitude	double	dst_latitude	double	dst_asn	bigint
dst_conn_type	varchar	dst_provider	varchar	dst_is_mobile	boolean
dst_is_cloud_provider	boolean	dst_is_proxy	boolean	dst_is_vpn	boolean
dst_is_tor	boolean	dst_is_tor_exit	boolean	hop_count	bigint
distance_km	double	src_prefix_8	varchar	src_prefix_16	varchar
dst_prefix_8	varchar	dst_prefix_16	varchar		

Dataset Characteristics & Exploratory Data Analysis (EDA)

Making Sense of the Dataset

The Generalized EDA

The following script performes some generailzed EDA like datatype/ missing value detection
between the target variable and the features.

In [3]:

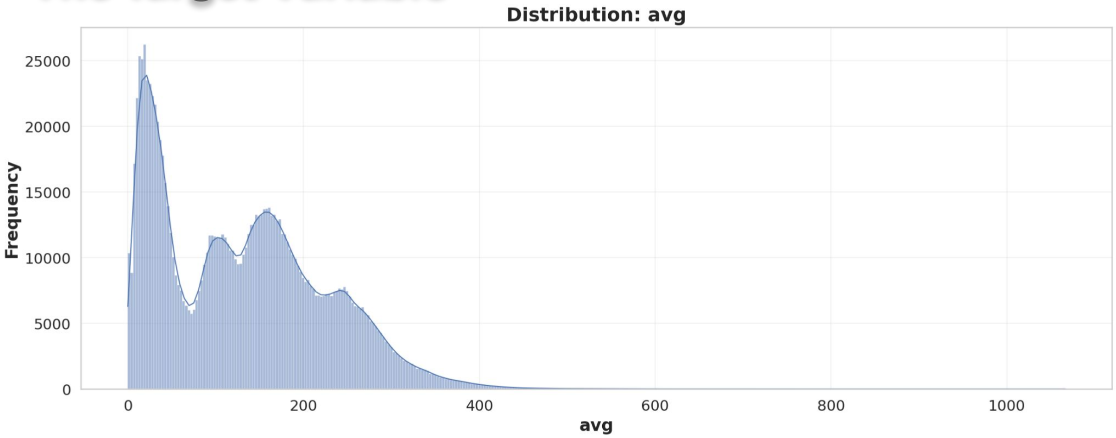
```
eda = EDAPipeline(df = df, target='avg')
eda.run_all()
```

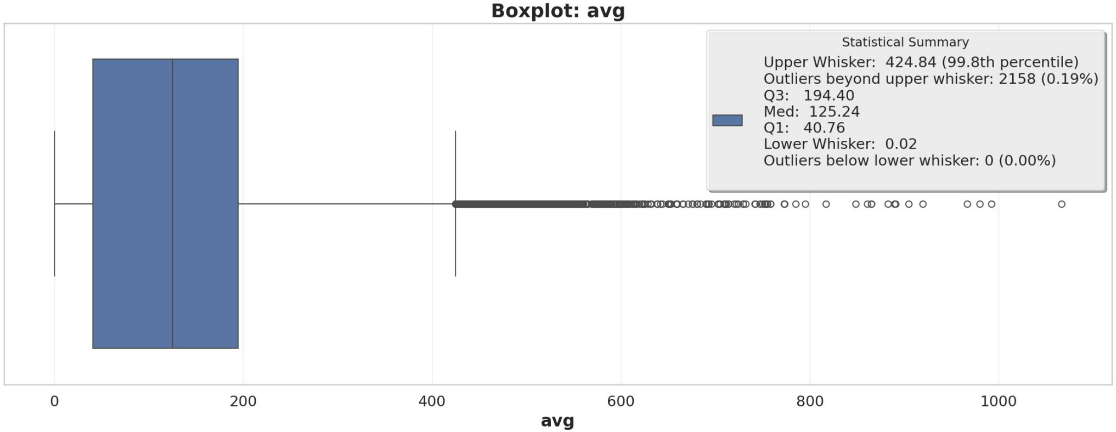
```
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
                    🚀 EDA Script for Initial Analysis Started !
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
============================================================
DATASET OVERVIEW
============================================================
Shape: (1116767, 41)
Number of rows: 1116767
Number of columns: 41
```

We have also -

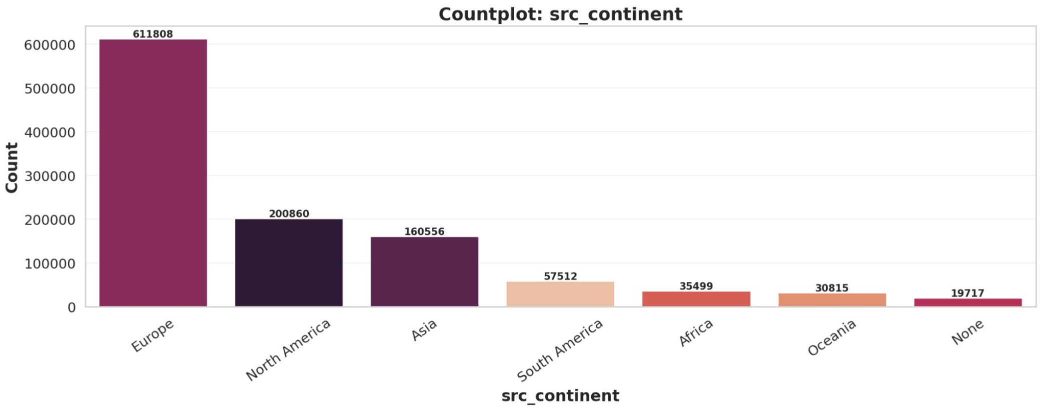
- created some domain specific features
- performed some targeted EDA

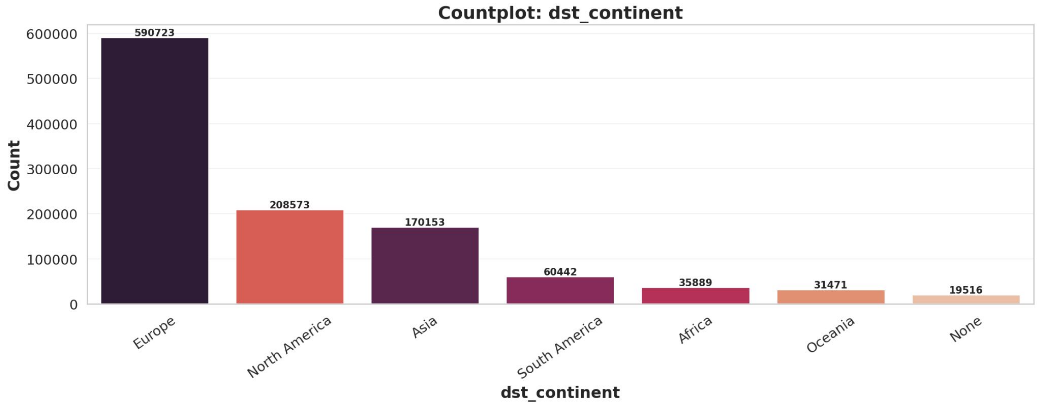
The Target Variable



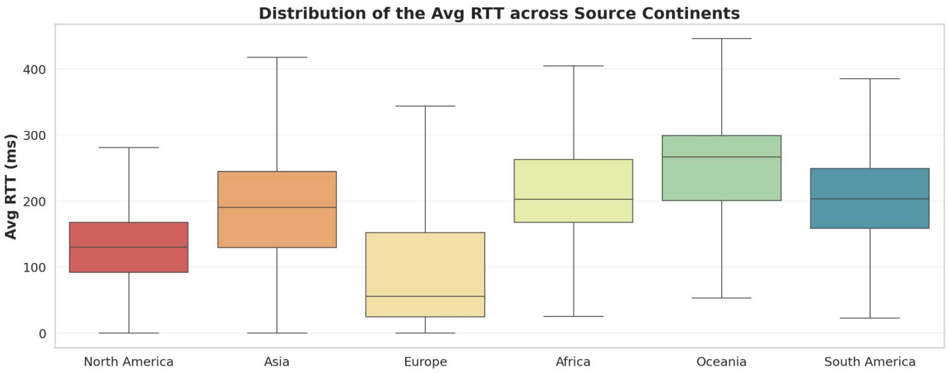


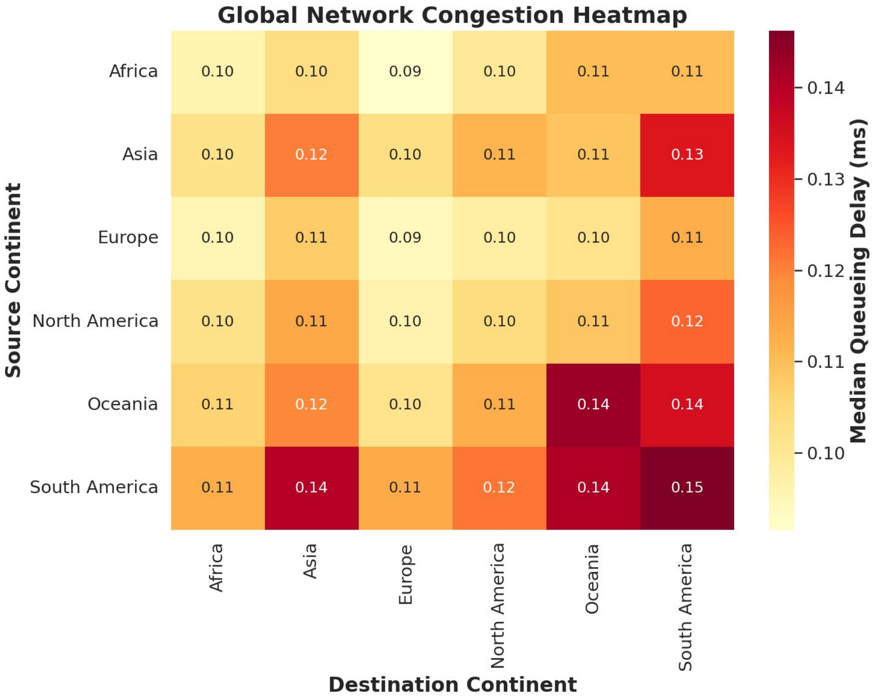
No Uniform Sampling





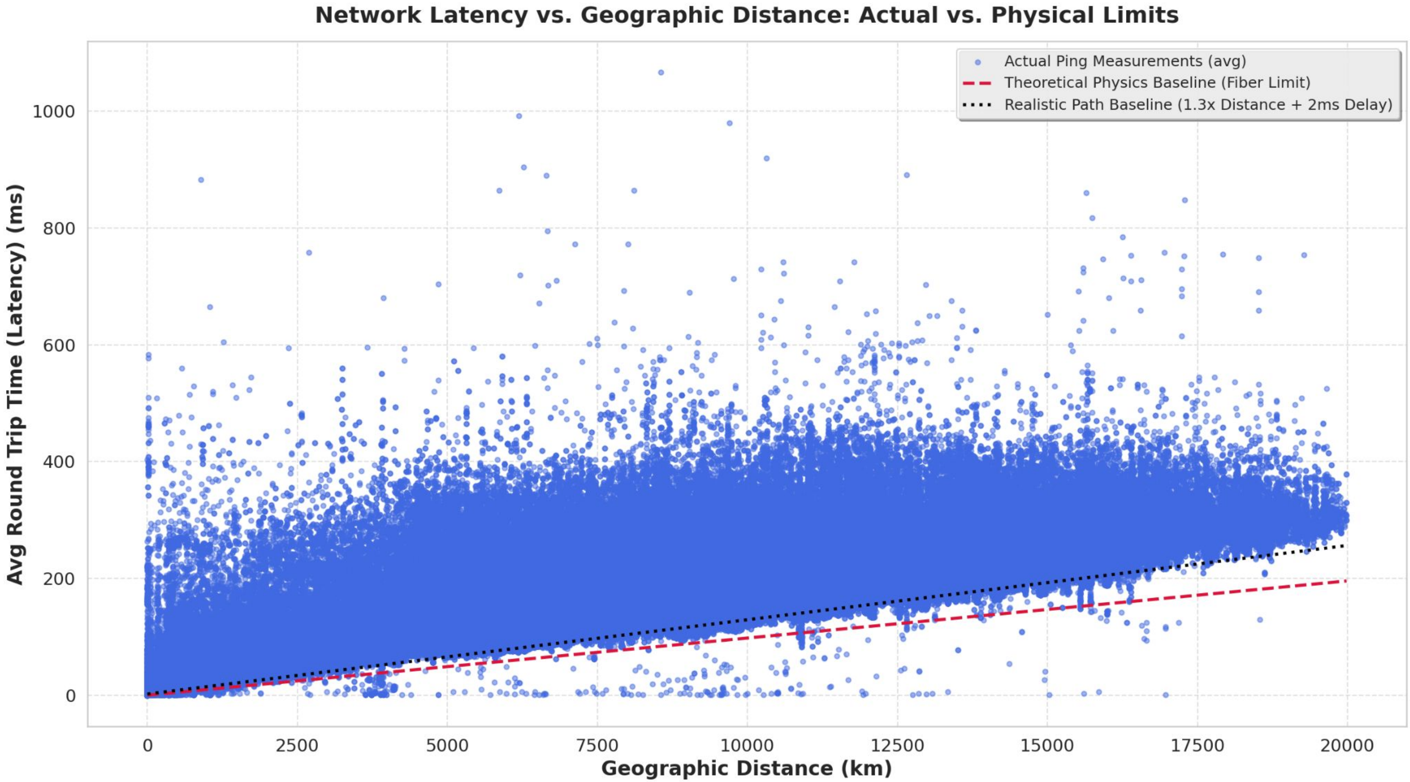
Quality of Service across Continents





Data Validation

Which measurements are valid ?



Models and Predictions

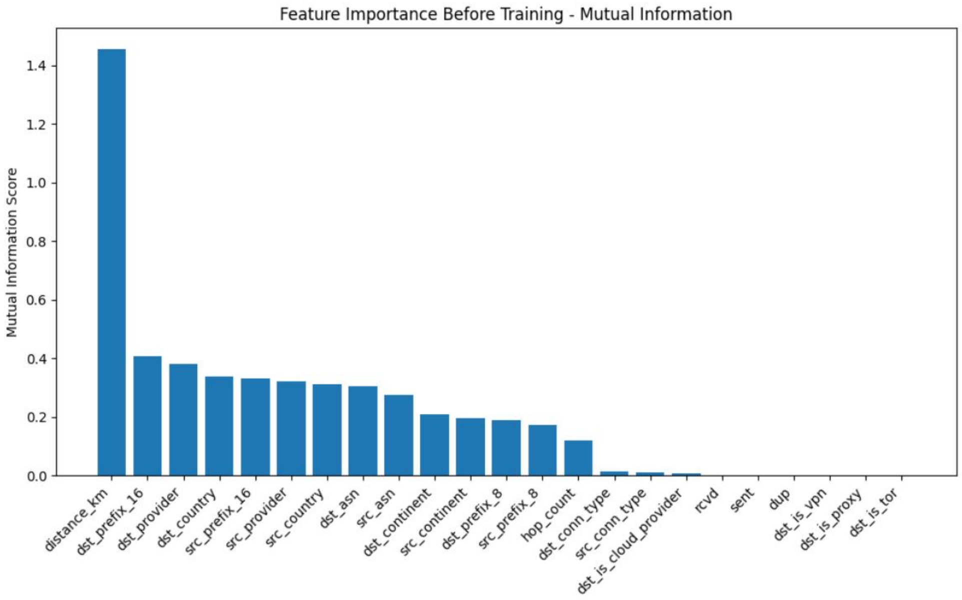
Feature Importance

Dataset

- 21 features

Feature Categories

Feature Category	Examples
Geographic	Distance, Country,Continent
Routing	ASN,Provider,Hop Count
Network	IP Prefixes, Connection Type



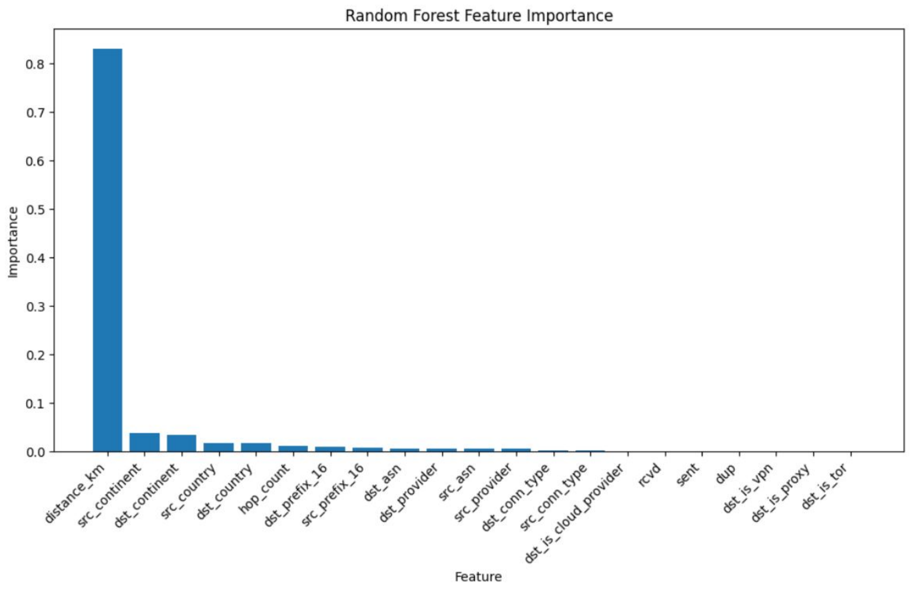
Baseline Model

Train/Test Split

- Random Split
- 80% Training - 20% Testing

Model

- Random Forest Regressor
- 100 trees
- Max Depth =20
- Min Samples Leaf =5



Model	MAE	RMSE	R^2
Random Forest	12.81 ms	22.95ms	0.939

Model Evaluation

Dataset

- Train/Test Dataset after Node Split
- 322k Training -62k Testing Samples

Model Comparison

Model	**Configuration**	**RMSE**	R^2
Random Forest	300 Trees • Depth=25	39.56 ms	0.829
XGBoost	300 Trees • Depth=8	37.46 ms	0.847
LightGBM	***500 Trees • 127 Leaves***	***36.96 ms***	***0.851***
MLP	256→128→64 •LR=10^{-3}	40.20 ms	0.824

LightGBM Analysis -I

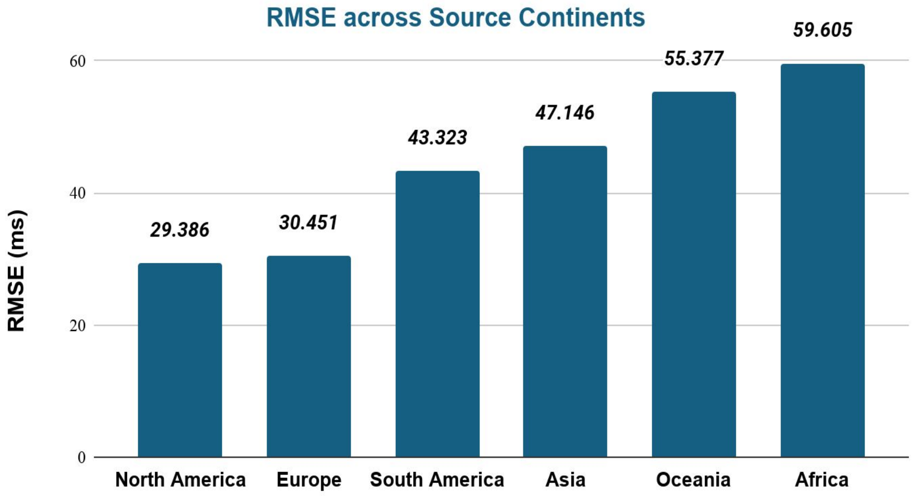
RTT Bucket (ms)	MAPE (%)
0-10	557.1
10-20	57.3
20-50	35.4
50-100	33.4
100-200	18.2
> 200	***12.4***

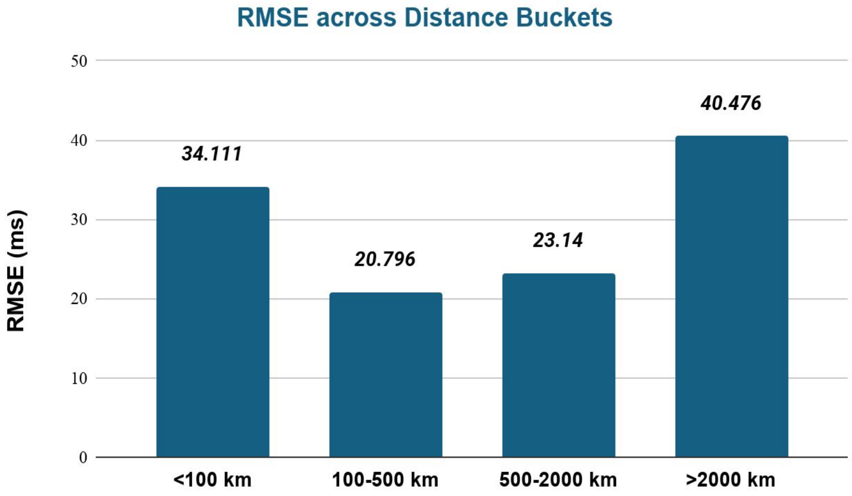
Source Continent	RMSE (ms)
North America	***29.35***
Europe	29.95
South America	40.85
Asia	48.00
Oceania	56.11
Africa	63.04

Distance (Km)	RMSE (ms)
100-500	***20.80***
500-2000	23.14
<100	34.11
>2000	40.48

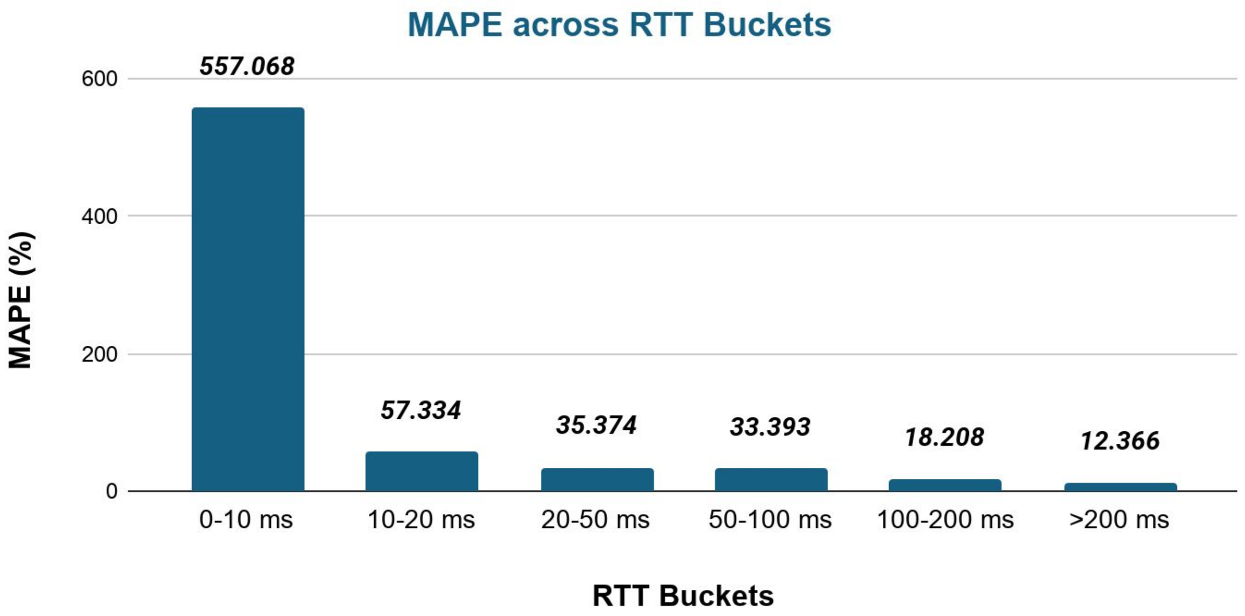
Percentile	Absolute Error (ms)
P50	16.66
P75	32.40
P90	***54.12***
P95	73.38
P99	134.60

LightGBM Model Performance -I





LightGBM Model Performance - II



Percentile	**Absolute Error (ms)**
P50	16.66
P75	32.40
P90	***54.12***
P95	73.38
P99	134.60

Conclusion & Future Direction

- LightGBM achieved the best overall prediction performance.
- Distance was the most informative predictor of RTT.
- Evaluate the model under more challenging deployment scenarios (e.g., ASN- or prefix-based splits).
- Explore Graph Neural Networks for network-aware RTT prediction.

Thank you for your attention!